Anonymisation Plan — ZW-BioBank v1.0

Zimbabwe Data Protection Act [Chapter 12:07] · structural controls first, masking second

Principle

The strongest anonymisation is structural: an identifier that is not in the dataset cannot leak from it. This plan therefore relies first on what the schema cannot store, and only second on masking what it can.

Direct identifiers

There is no field in this schema for a holder's name, address, telephone number or national identity number. Holders are represented by a pseudonym (holder_pseudonym_id). The register mapping pseudonyms to people is AES-256 encrypted, held in SIRDC institutional custody, and distributed at no access tier — not to research partners, not to commercial licensees, not to international collaborators. Keeping that register outside the dataset is what makes the rest of the dataset shareable at all. It is also what makes benefit-sharing enforceable: the link survives, under Zimbabwean custody, for as long as it is needed.

The residual risk is free text. A holder's name can arrive inside a verbatim description. The validator scans free-text fields on every ingest for telephone and national ID patterns and raises a critical failure on a match, because a pseudonym is worth nothing if the name is written in the next column.

Location

Tier	Coordinate treatment	Stated precision
Public	Ward centroid substituted for the field position; location_masked = 1	5,000 m
Research	Ward centroid, unless the research question demonstrably requires finer resolution and SIRDC approves	5,000 m
Restricted	Field GPS as captured	5 m
Commercial	Ward centroid. Finer resolution is not licensable.	5,000 m

The reason is not abstract. A public dataset giving field-precision coordinates for a medicinal plant that a community depends on is a harvest map. *Warburgia salutaris* is already under pressure from bark stripping across its range. Publishing exactly where the remaining trees stand would cause the harm this project exists to prevent, and would do it faster than any benefit the dataset delivered.

The schema enforces the rule: **CHECK (location_masked = 0 OR gps_precision_m >= 1000)**. A record cannot claim to be masked while carrying field-precision coordinates. The validator additionally fails any public-tier record that is not masked.

Attribution

Attribution defaults to the community, identified by district and ward. Naming an individual holder requires an explicit opt-in recorded in named_attribution_opt_in, captured at consent. Nobody is named by default, and nobody is named retrospectively.

Withdrawal

On receipt of a withdrawal, view v_withdrawal_impact enumerates every downstream sample, sequence and metabolite record. Each is pseudonymised or deleted within 30 days. Until they are, the validator raises a critical failure on every run and the batch is held. Withdrawal is therefore not a task someone can forget: the pipeline stops until it is done.

Re-identification risk

The residual risk is inference rather than disclosure. A ward with a single known practitioner, combined with a distinctive plant and a specific indication, could narrow identity to one person even with the name removed. Two controls apply: heritage records classified community-restricted are not released to the research tier at all, and any aggregate release suppressing fewer than five holders in a ward is withheld. This risk is disclosed rather than claimed to be eliminated, because it cannot be eliminated while the data remains useful.